Project-1

Mobile Tech Retailers Inc. | Cellphone Information Database

1. Company Background

MobileTech Retailers Inc. is a leading nationwide cell phone distribution and retail chain with a presence across multiple US states. Founded in the early 2000s, MobileTech partners with top global manufacturers — including Samsung, Apple, Nokia, and Motorola — to offer a wide range of handsets catering to every customer segment, from budget-conscious buyers to premium smartphone enthusiasts.

With thousands of transactions processed each month across diverse geographies, MobileTech operates a centralised data warehouse to consolidate all sales activity. This enables the leadership team to make fast, evidence-based decisions on inventory, pricing, partnerships, and customer engagement.

2. Business Scenario

The Sales & Strategy team at MobileTech is preparing for its annual business review. To support the review, the Head of Analytics has tasked the data team with producing a comprehensive analysis of the company's transaction history stored in the Cellphones Information database.

The analysis must address questions across four key business domains:

- Geographic Performance — Understanding which markets drive sales and where growth opportunities lie
- Product & Pricing Intelligence — Identifying bestsellers, pricing benchmarks, and manufacturer competitiveness
- Customer Behaviour — Profiling high-value customers, spending trends, and year-on-year changes
- Manufacturer & Supply Chain Insights — Evaluating manufacturer performance, new entrants, and strategic partnerships

As the lead Data Analyst assigned to this project, you are required to write SQL queries against the Cellphones Information database to answer 10 business questions raised by senior stakeholders.

3. Database Schema — Cellphones Information

The database follows a Star Schema design with one central Fact table recording every sales transaction, supported by four Dimension tables that provide contextual details about customers, products, locations, and dates.

DIM_MANUFACTURER — Stores details of all phone manufacturers

Column Name	Data Type	Description
IdManufacturer	INT	Primary Key — unique manufacturer identifier
Manufacturer_Name	VARCHAR(20)	Name of the manufacturer (e.g., Samsung, Apple)

DIM_MODEL — Stores details of all phone models

Column Name	Data Type	Description
IdModel	INT	Primary Key — unique model identifier
Model_Name	VARCHAR(20)	Commercial name of the phone model
Unit_Price	MONEY	Retail unit price of the model
IdManufacturer	INT	Foreign Key — links to DIM_MANUFACTURER

DIM_CUSTOMER — Stores customer profile information

Column Name	Data Type	Description
IdCustomer	INT	Primary Key — unique customer identifier
Customer_Name	VARCHAR(20)	Full name of the customer
Email	VARCHAR(100)	Customer email address
Phone	INT	Customer contact number

DIM_LOCATION — Stores geographic information for each transaction

Column Name	Data Type	Description
IdLocation	INT	Primary Key — unique location identifier
ZipCode	INT	Postal zip code
Country	VARCHAR(20)	Country name
State	VARCHAR(20)	US state name
City	VARCHAR(20)	City name

DIM_DATE — Stores date dimension for time-based analysis		
Column Name	**Data Type**	**Description**
Date	`DATE`	Primary Key — full calendar date
Year	`INT`	Calendar year
Quarter	`VARCHAR(20)`	Quarter label (Q1, Q2, Q3, Q4)
Month	`VARCHAR(20)`	Month name or number

FACT_TRANSACTIONS — Central fact table recording every sales transaction		
Column Name	**Data Type**	**Description**
IdModel	`INT`	Foreign Key — phone model sold
IdCustomer	`INT`	Foreign Key — customer who made the purchase
IdLocation	`INT`	Foreign Key — location of the transaction
Date	`DATE`	Foreign Key — date of the transaction
TotalPrice	`MONEY`	Total revenue from the transaction
Quantity	`INT`	Number of units purchased

4. Business Questions

The following 10 questions have been raised by MobileTech's senior leadership. Each question is accompanied by its business objective to help you understand the context and frame your analysis accordingly.

Q#	Business Question	Business Objective
Q1	Which states have active customers who have purchased cellphones since 2005?	*Identify geographic markets where MobileTech has had sustained sales presence to guide regional expansion strategy.*
Q2	Which US state records the highest volume of Samsung cellphone sales?	*Pinpoint Samsung's strongest regional market to optimize inventory allocation and targeted promotions.*
Q3	How many transactions have been made for each phone model, broken down by zip code and state?	*Enable hyper-local demand planning so the supply chain team can stock the right models in the right locations.*
Q4	Which cellphone model is the most affordable, and what is its unit price?	*Understand the entry-level price point in the portfolio to position budget-friendly campaigns for price-sensitive customers.*

Q5	What is the average selling price per model among the top 5 manufacturers by total units sold?	*Benchmark pricing across high-volume manufacturers to assess margin health and identify premium vs. volume-driven brands.*
Q6	Which customers had an average spend above $500 in the year 2009, and what was their average spend?	*Identify high-value customer segments from 2009 for loyalty programme targeting and personalized retention offers.*
Q7	Were there any phone models that consistently ranked in the top 5 by units sold across 2008, 2009, and 2010?	*Discover evergreen bestsellers that maintained sustained demand over three consecutive years to anchor procurement decisions.*
Q8	Which manufacturer ranked second in total sales revenue in 2009, and which ranked second in 2010?	*Track shifts in the competitive manufacturer landscape year-over-year to renegotiate supplier agreements strategically.*
Q9	Which manufacturers were active in 2010 but had no recorded sales in 2009?	*Flag newly onboarded or re-entering manufacturers in 2010 to evaluate their market contribution and partnership potential.*
Q10	For the top 100 customers by lifetime spend, what is their average annual spend, average quantity per year, and how has their spend changed year-over-year as a percentage?	*Monitor the spending trajectory of MobileTech's most valuable customers to proactively detect churn risk or upsell opportunities.*